SLEEVEDRIFT · PROJECT PLAN

SleeveDrift project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue at the kit 1.9.1 upgrade; stage P3 reached, P4 to P6 on hold under the stage cap

SleeveDrift project plan

SleeveDrift is at stage P3 (TRL 3, a constructable design with a detailed model of every concept option). The next gate is Amish's choice between options A and B (SVD-OPT-001, within 10 % on score) and his answer on the cost target (PR-013); after that the work is hands-on and waits for the portfolio's TRL cap to lift. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Project lead	SVD-PRB-001, SVD-URS-001	Done at P3
STEM check and calculations	Design lead	SVD-STM-001, SVD-CAL-001 (sizing.py, stem_check.py)	Done; independent review recorded
Concept options and selection	Design lead	SVD-OPT-001, SVD-DDR-005	Done; decision awaiting Amish
Requirements and traceability	Design lead	SVD-PRS-001, SVD-RTM-001 (generated)	Done
Detailed CAD of every option	CAD lead	cad/src/model.py, cad/fidelity.json, STEP, drawings	Done at D3
Build plan	Design lead	SVD-BLD-001 and its pictures	Done (paper)
Regulatory, feasibility, risk, market	Project lead	SVD-REG-001, SVD-FEA-001, SVD-RSK-001, SVD-MKT-001	First issue
Co-design partner	Project lead	An agreed partner and review of the user requirements	Not started; NDRF first (not agreed)
Hardware build and trials	Test lead	Prototype, fit trial, load tests, timed trials	On hold (P4 to P6 under the stage cap)

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design partner time	Review of URS and PRS; site pipe data	A few days	to assess
Steel workshop with plate rolls	Prototype ring, sledges, trestles (made parts)	About 3 weeks	1,637 in parts for the made lines; labour to assess
Bought items	Winches, drills, monitors, blower, duct, ropes	One order	9,167 (the rest of the BOM), first-pass prices
CalRig or equivalent	Hood and ring proof loads	Days	to assess
Pipe test section, 800 x 10 mm	Fit, ring friction and haul trials	12 m or more	to assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 to P2	Problem, concept, first calculations	Done (TRL 2, 2026-10-03)
P3	Constructable design, detailed model of every option, STEM check, options, requirements traced, build plan	Reached; awaiting Amish on the concept choice
P4	Prototype built; fit trial, ring friction, hood load, haul and set-up timed	On hold (stage cap)
P5	Endurance and field trial with a partner	On hold
P6	Release candidate; certification routes assessed	On hold

Next steps and decision requested

1. **Decision for Amish:** the concept (SVD-DDR-005). Option A scores 175 of 195, B 164 and C 148; A and B are within 10 %, so both are put to Amish without a pick. A is the current design and the baseline.
2. **Decision for Amish:** hold or restate the USD 5,000 cost target (PR-013); the parts cost is USD 10,804.00 (SVD-DEC-001 open decision 2).
3. **Decision for Amish:** the jacking screw length, M20 x 60 as modelled or the 90 mm in the BOM (SVD-DEC-001 open decision 3).
4. Approach the first co-design partner (SVD-PRB-001, Co-design) once the choice is made.
5. When the cap lifts: TRL 4 starts with the pull-out test of the ring, the swivel release batch, the sled test and the 12 m fit trial (the highest RPNs in SVD-RSK-001).

References

- OSHA, "Permit Required Confined Spaces", 59 FR 60735, https://www.osha.gov/node/16671, accessed 2026-11-02.
- FireRescue1, "NFPA 1983: Where did it come from and what does it mean?", https://www.firerescue1.com/fire-products/technical-rescue/nfpa-1983-where-did-it-come-from-and-what-does-it-mean, accessed 2026-11-02.

- European Agency for Safety and Health at Work, "Regulation (EU) 2023/1230 on machinery", https://osha.europa.eu/en/legislation/directive/regulation-20231230eu-machinery, accessed 2026-11-02.
- De Gemmill, "Frontline Megapod ...", https://degemmill.com/products/frontline-megapod-confined-space-7-ft-aluminum-tripod-with-60-ft-winch-and-3-way-srl, accessed 2026-11-02.
- Hazchem Safety, "BW Gas Alert Micro Clip XL 4 Gas Detector", https://www.hazchemsafety.com/product/bw-gas-alert-micro-clip-xl-4-gas-detector-o2-lel-co-h2s/, accessed 2026-11-02.
- Silkyara sources: see SVD-PRB-001.